

Sourabh Guduru

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EDUCATION

Carnegie Mellon University , Pittsburgh, PA, USA	December 2023
Master of Science with focus in Data Science (CHE)	
Visvesvaraya National Institute of Technology , Nagpur, Maharashtra, India	May 2022
Bachelor of Technology in Engineering	

WORK EXPERIENCE

Data Scientist – Research Assistant, Carnegie Mellon University , Pittsburgh, USA	Jan 2023 – Dec 2023
- Developed computationally efficient predictive model to enhance climate change projections, reducing prediction uncertainty by 60% and improving runtime by optimizing key model parameters. - Implemented data processing and transformation workflows to ensure efficient feature engineering and validate model assumptions, using statistical tests (hypothesis testing, normality checks) - Developed KPIs for climate projections, visualized with Tableau, and applied Bayesian modeling for parameter optimization, reducing model execution time and improve accuracy.	
Data Scientist Intern, Turilytix.AI , Remote, USA	May 2023 - Aug 2023
- Drove \$1.5 million cost reduction by developing ML Pipelines for predictive maintenance on imbalanced IoT data. - Improved model performance metrics (F1 score from 0.52 to 0.78) using XGBoost on imbalanced IoT data and selected relevant features through ANOVA and Chi-Square tests. - Designed and implemented the end-to-end solution, from data preprocessing to deployment, ensuring seamless integration of predictive maintenance ML pipelines. - Utilized regression analysis and time series techniques, to quantify the impact of sensor fluctuations on failure prediction, leading to 20% reduction in downtime. - Presenting model outcomes to senior management, earning commendations for solution structure, timely delivery.	
Research Assistant, VNIT , Nagpur, India	Aug 2021 – May 2022
- Performed causal analysis to optimize efficiency through IoT sensor data analysis with statistical techniques and ML. - Conducted Root Cause Analysis on anomalies, leading to an impressive 82% efficiency improvement.	
Business Analyst Intern, Increff , Bangalore, India	Jan 2022 - Feb 2022
- Drove 6% revenue growth for top retail and e-commerce brands through data-driven inventory optimization and supply chain efficiency improvements. - Achieved a 10% decrease in inventory holding, resulting in significant cost savings. Created Tableau dashboards to aid decision making to improve bottom-line and reduce overhead costs by predictive analytics models.	

PROJECTS

AI Medical Chatbot

- Developed a medical chatbot in Flask, using a quantized Llama LLM with prompt engineering for improved response accuracy, integrated LangChain with RAG for semantic search, and utilized Pinecone for efficient data management.
- Achieved a BERTscore of 0.82 for generated response indicating a good alignment with the referenced text.

Dynamic Stock Investment Strategy

- Implemented predictive modeling techniques to navigate the volatile NASDAQ market for stock investment strategy.
- Applied ARIMA and Random Forest for stock price forecasting and profitability classification, ensuring a comprehensive approach to investment decision-making. Analyzed NASDAQ data, resulting in a cumulative profit of approximately 14% over 3 months.

Identifying Streaming Consumers through ML Classification

- Delivered impactful insights into streaming service consumer behavior using Random Forest model. Performed ETL process on media survey data obtained through an API and stored in AWS S3, driving targeted marketing strategies.
- Streamlined the model, reaching an F1 score of 81%, which enhanced client marketing strategies and lowered acquisition costs with better targeted adds.

Fraud Detection and Risk Assessment Model

- Built Fraud Detection model using XGBoost to identify and classify fraudulent transactions, achieving F1 score of 78%.
- Employed techniques like SMOTE for balancing datasets and performed feature engineering to extract key fraud indicators, leading to 27% improvement in identifying high risk transactions.

SKILLS

Languages: Python, SQL, R.

Libraries/Frameworks: Pandas, NumPy, Spark, Scikit-learn, Matplotlib, Seaborn, PyTorch, Keras, TensorFlow.

Cloud/Data Analysis: Azure, Looker, Excel, Git, AWS (S3, EC2), Hadoop, SQL, PostgreSQL, Jupyter, Google Colab.

Data Science: A/B Testing, Gradient Boosting, Statistics, SAS, Deep Learning, NLP, Machine Learning, K-Nearest Neighbors, Support vector Machines, Linear Regression, Logistic Regression, Decision Trees, K-Means, Data Visualization, Risk Analytics.

Functional: Communication, Cross-Functional, Problem-Solving, Business/ Technical Recommendations, Business Intelligence.